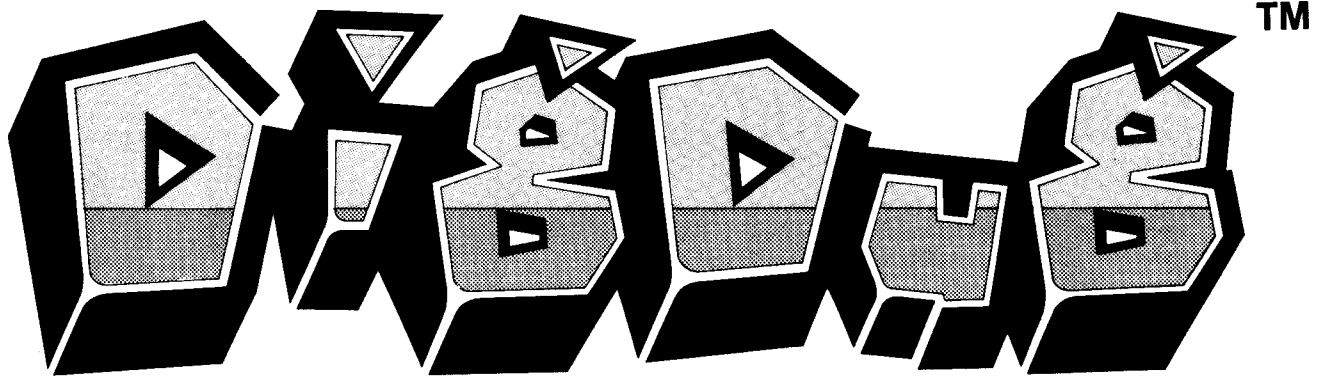
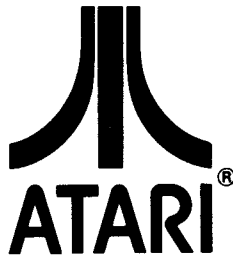


Schematic Package Supplement to



Operation, Maintenance and Service Manual



 A Warner Communications Company

© ATARI INC. 1982

Table of Contents

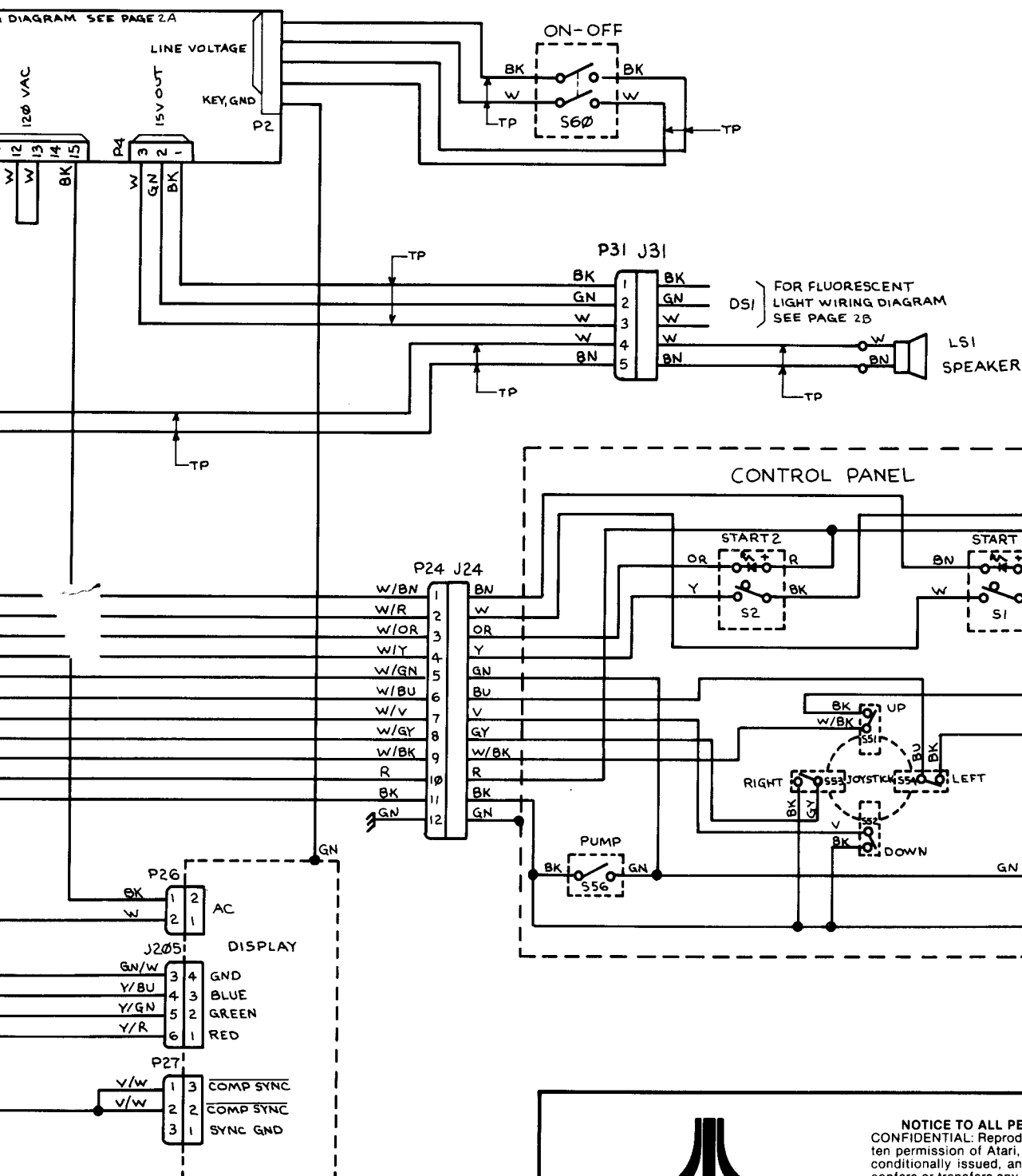
Sheet 1A	You Are Here
Sheet 1B	Dig Dug Upright Wiring Diagram (038294-01 A)
Sheet 2A	Color Raster-Scan Power Supply Wiring Diagram (037669-01 C) Regulator/Audio II PCB Schematic Diagram (035435-05 A)
Sheet 2B	Coin Door Wiring Diagram (037542-01 A) Utility Panel Wiring Diagram (038004-01 A) Fluorescent Light and Speaker Wiring Diagram (035833-01 A)

Game PCB Schematics, Sheets 3A—8A

Sheet 3A	CPU PCB Power Input, Clock, NMI, Watchdog, and Memory Map
Sheet 3B	1st Priority CPU, 2nd Priority CPU
Sheet 4A	3rd Priority CPU, CPU PCB Address Decoder
Sheet 4B	Switch Inputs, Coin Door and Control Panel Outputs
Sheet 5A	Audio Output
Sheet 5B	Sync Generator
Sheet 6A	Playfield Generator
Sheet 6B	Motion Object RAM, Motion Object Address Generator, Decoder and Match Line Flag
Sheet 7A	Motion Object Generator
Sheet 7B	Video Output, Video Color Controller
Sheet 8A	High-Score Table
Sheet 8B	Electrohome 19-Inch Color Raster-Scan Video Display Schematic Diagram

NOTE
This staple temporarily holds the schematic package together. Remove the staple before using the schematics.

DIAGRAM SEE PAGE 2A



NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
 CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

Dig Dug Game Wiring Diagram

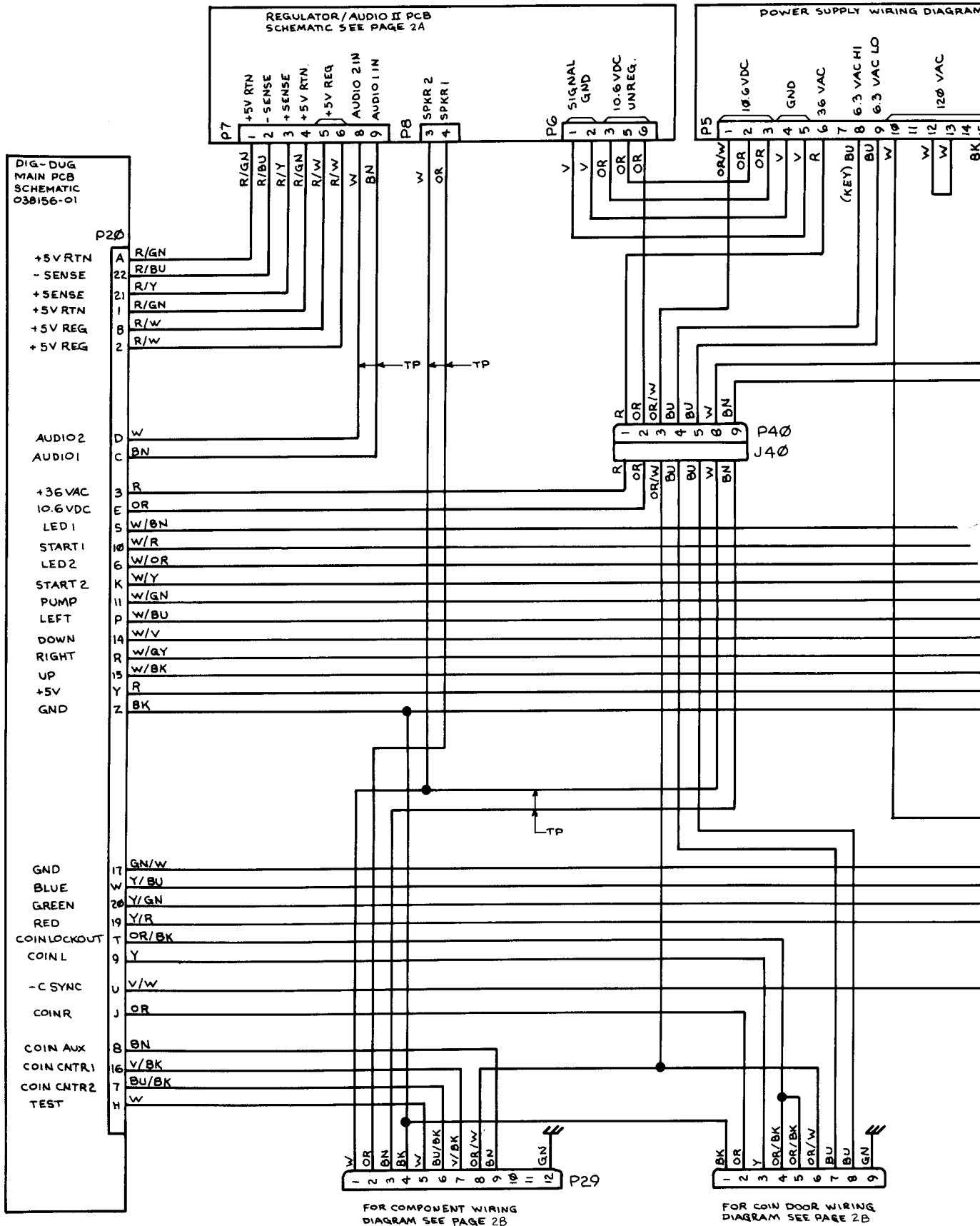
© ATARI INC., 1982



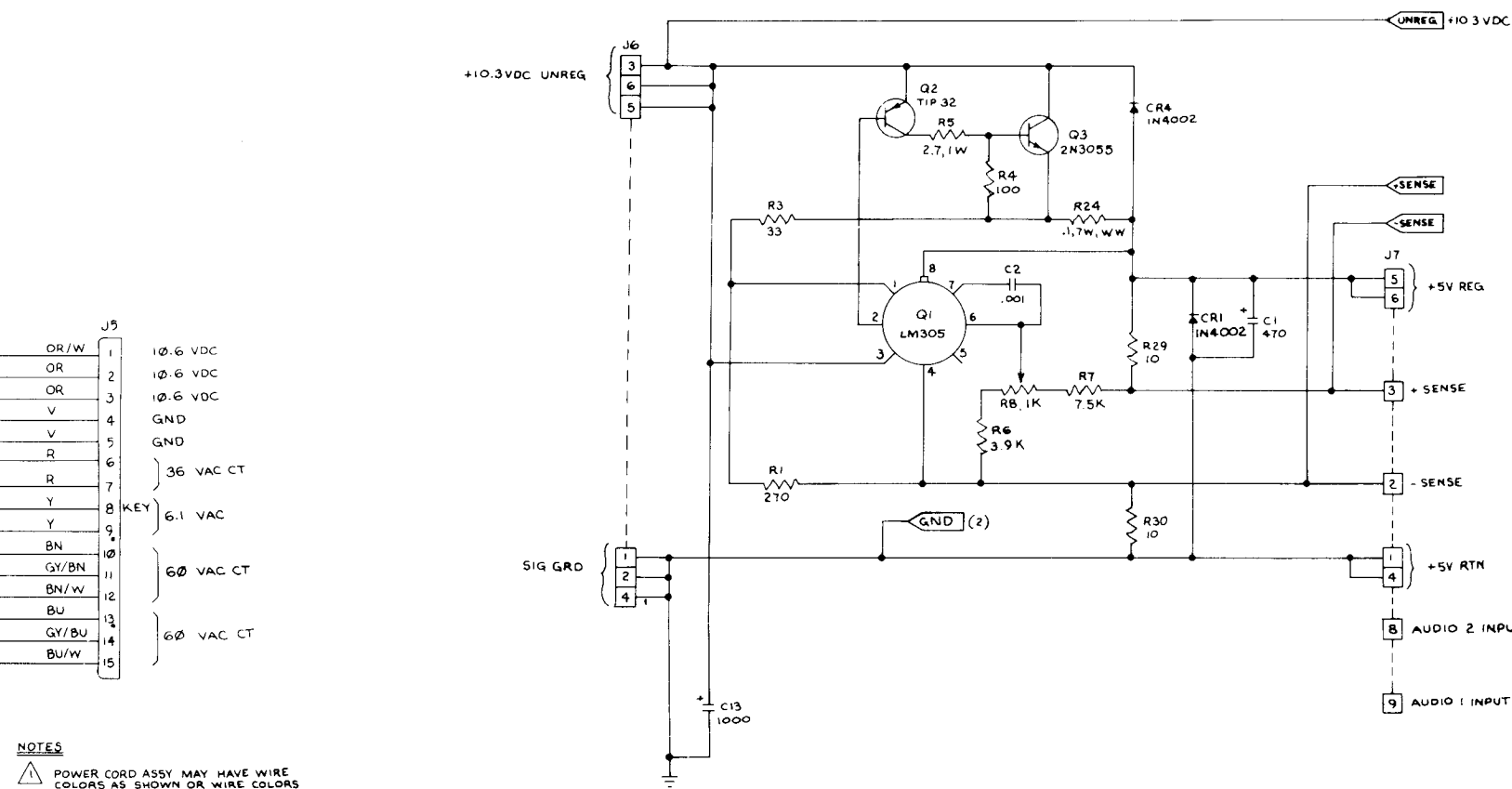
A Warner Communications Company

SP-203 Sheet 1B
 1st printing 3L

Dig Dug Upright Game Wiring Diagram



Regulator/Audio II PCB Schematic Diagram



NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
 CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

Dig Dug Power Supply and Reg./Audio II PCB

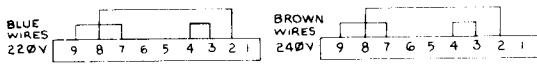
© ATARI INC., 1982



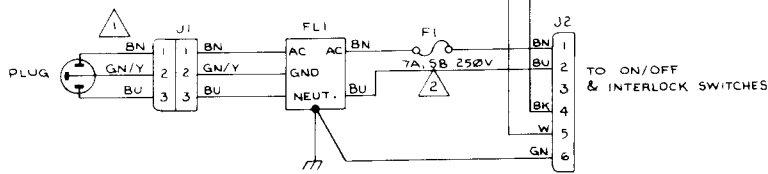
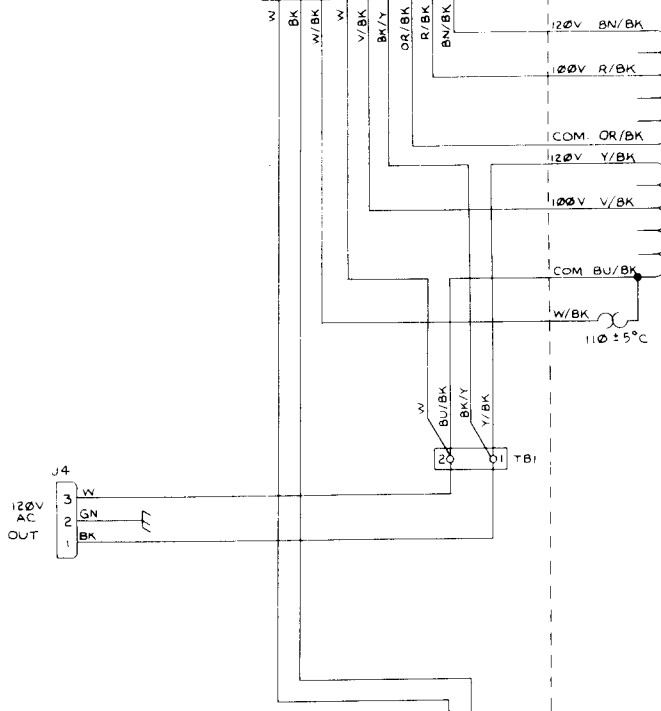
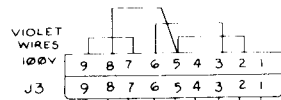
A Warner Communications Company

SP-203 Sheet 2A
 1st printing 3L

Color Raster Power Supply Wiring Diagram



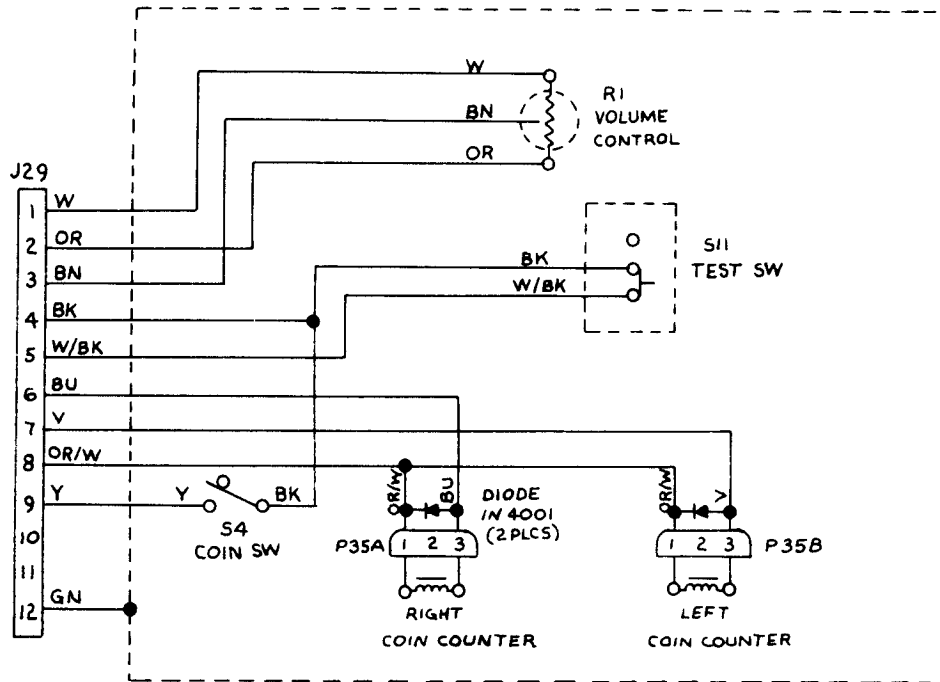
VOLTAGE SELECTION BLOCKS



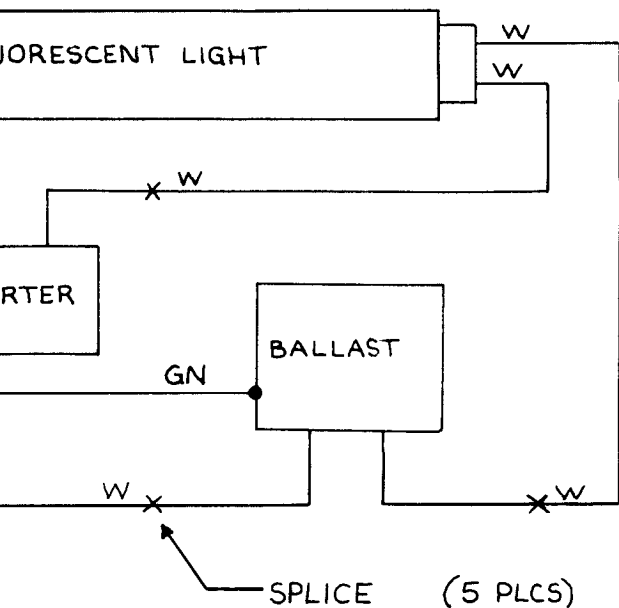
NOTES

- △ PO
- △ CO
- △ A:
- △ O:
- △ W:
- △ US
- △ 2:

Utility Panel Wiring Diagram



And Speaker Wiring Diagram



NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
 CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

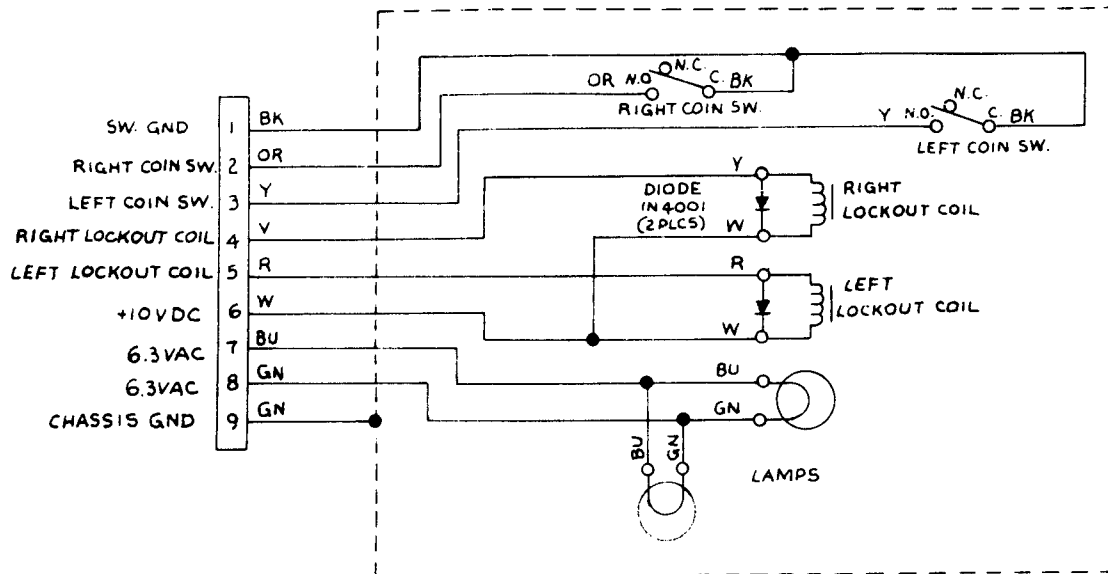
Dig Dug Game Wiring Interfaces

© ATARI INC., 1982

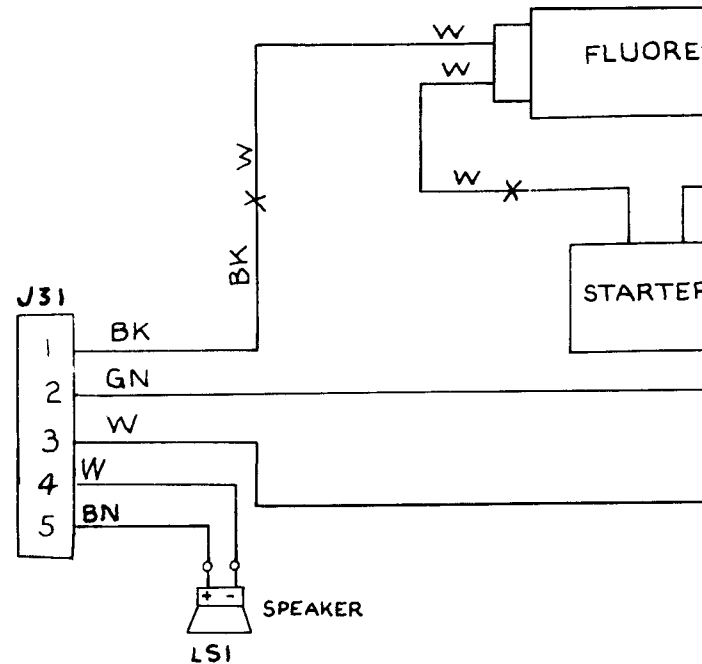
A Warner Communications Company

SP-203 Sheet 2B
 1st printing 3L

Coin Door Wiring Diagram

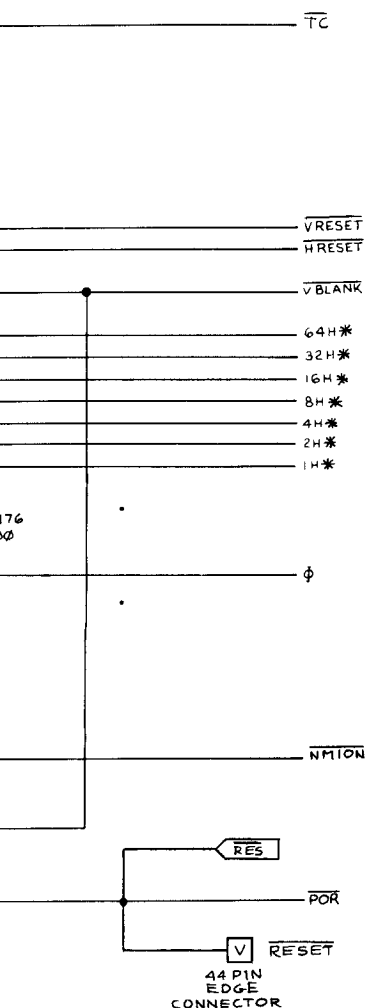


Fluorescent Light And



MEMORY MAP

HEXA- DECIMAL ADDRESS	R/W	DATA								FUNCTION
		D7	D6	D5	D4	D3	D2	D1	D0	
0000-3FFF	R	D	D	D	D	D	D	D	D	1st Priority Z80 CPU ROM (16K)
0000-1FFF	R	D	D	D	D	D	D	D	D	2nd Priority Z80 CPU ROM (8K)
0000-0FFF	R	D	D	D	D	D	D	D	D	3rd Priority Z80 CPU ROM (4K)
6800-680F	W					D	D	D	D	Audio Control
6810-681F	W					D	D	D	D	Audio Control
6820	W								D	0 = Reset IRQ1 (Latched)
6821	W								D	0 = Reset IRQ2 (Latched)
6822	W								D	0 = Enable NMI3 (Latched)
6823	W								D	0 = Reset 2nd and 3rd Z80 CPUs (Latched)
6825	W								D	Custom Chip 53 Mode Control (Latched)
6826	W								D	Custom Chip 53 Mode Control (Latched)
6827	W								D	Custom Chip 53 Mode Control (Latched)
6830	W									Watchdog Reset
7000	R/W	D	D	D	D	D	D	D	D	Custom Chip 06—Data
7100	R/W	D	D	D	D	D	D	D	D	Custom Chip 06—Command
8000-87FF	R/W	D	D	D	D	D	D	D		2K Playfield RAM
8800-8BFF	R/W	D	D	D	D	D	D	D	D	1K Motion RAM (PIC)
9000-93FF	R/W	D	D	D	D	D	D	D	D	1K Motion RAM (HPOS, VPOS)
9800-9BFF	R/W	D	D	D	D	D	D	D	D	1K Motion RAM (FLIP)
A000	W								D	Playfield Select (Latched)
A001	W								D	Playfield Select (Latched)
A002	W								D	Playfield Color Select (Latched)
A003	W								D	Alphanumeric Color Select (Latched)
A004	W								D	Playfield Color Select (Latched)
A005	W								D	Playfield Color Select (Latched)
A007	W								D	Flip Video
B800-B83F	W	D	D	D	D	D	D	D	D	Write EAROM Address and Data
B800	R	D	D	D	D	D	D	D	D	Read EAROM Data
B840	W					D	D	D	D	Write EAROM Control



NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
 CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

Dig Dug CPU PCB Schematic Diagram

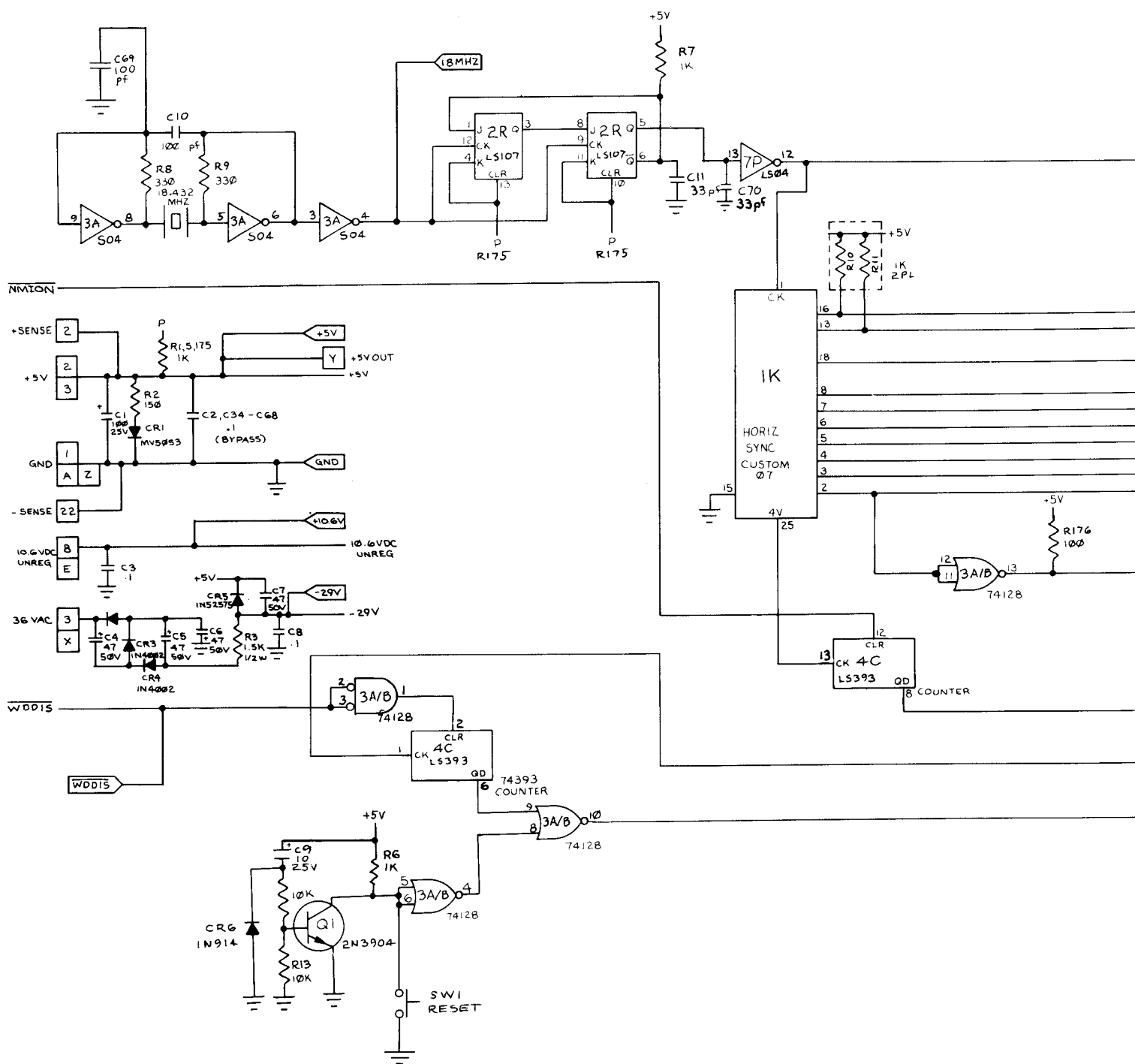
© ATARI INC., 1982



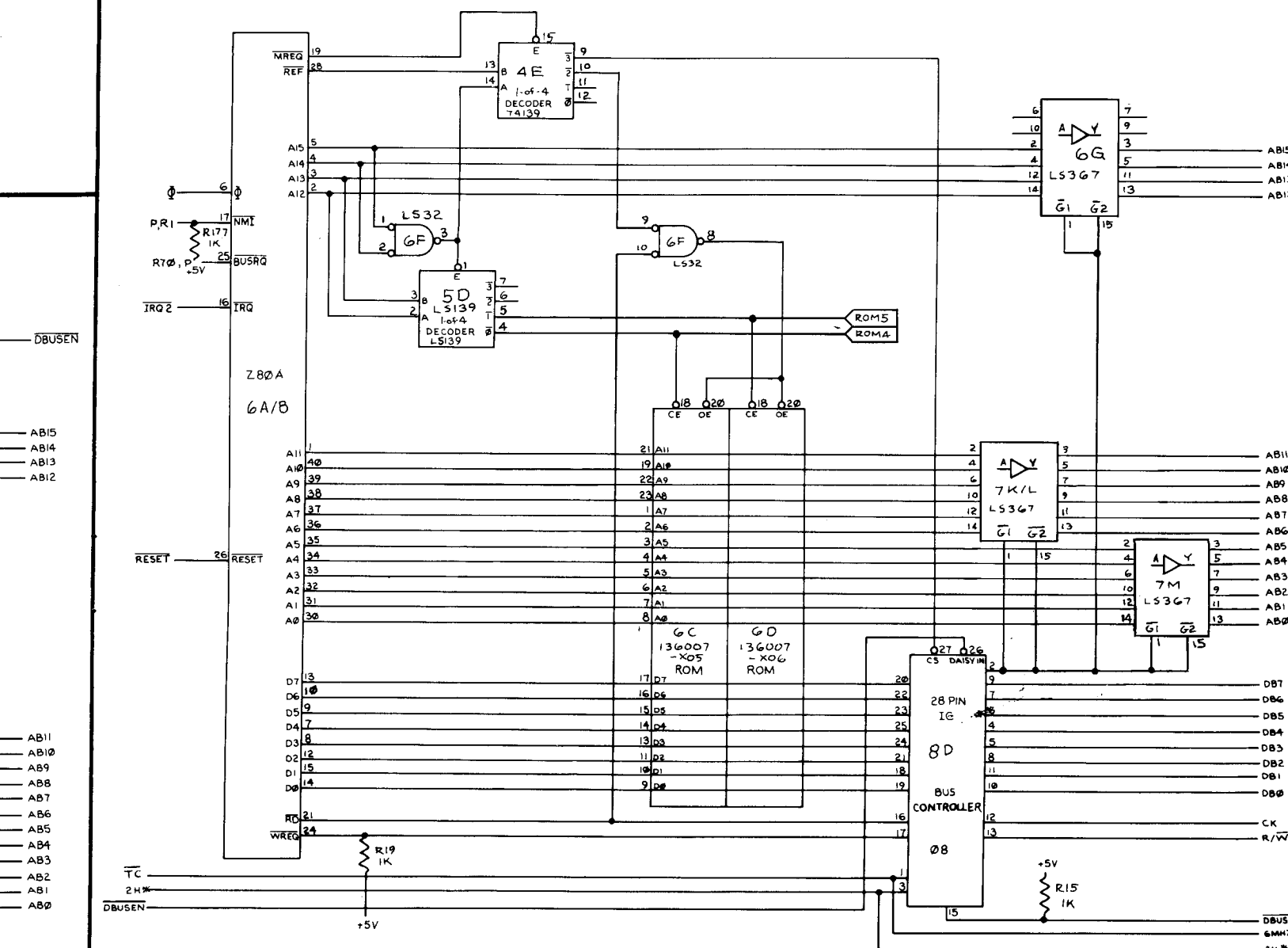
A Warner Communications Company

SP-203 Sheet 3A
 1st printing 3L

CPU PCB Power Input, Clock, NMI, and Watchdog



2nd Priority CPU



NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
 CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

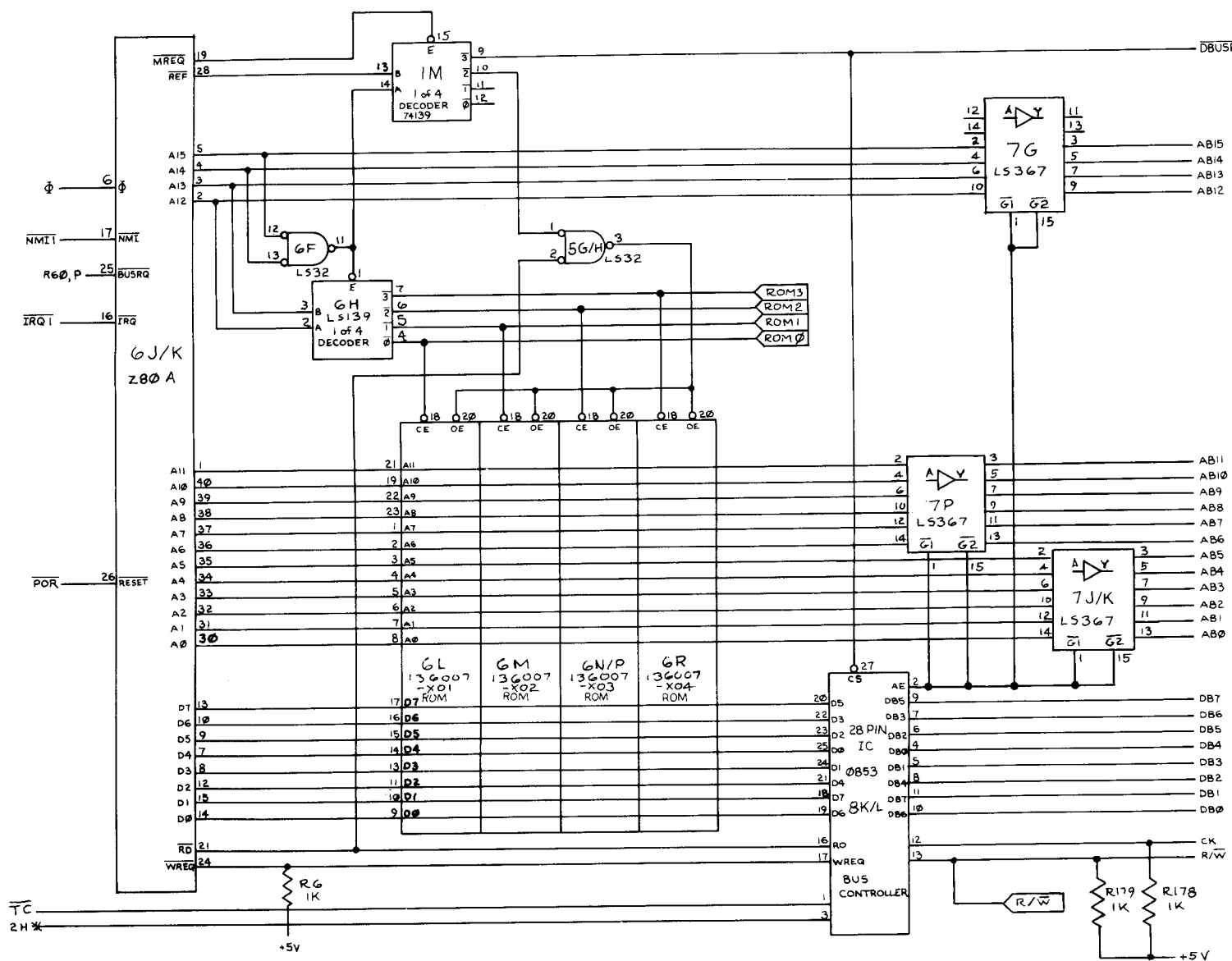
Dig Dug CPU PCB Schematic Diagram

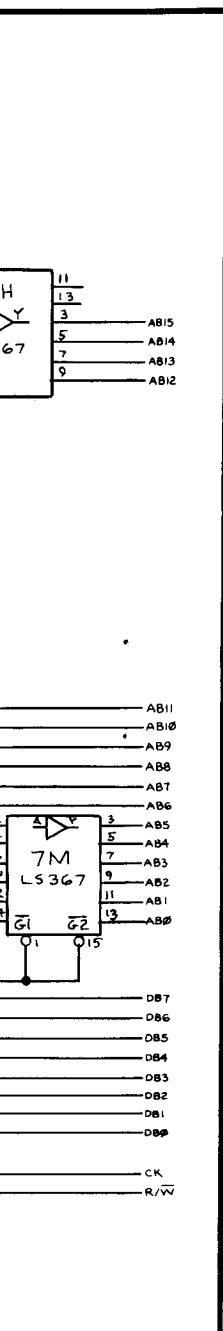
© ATARI INC., 1982

A Warner Communications Company

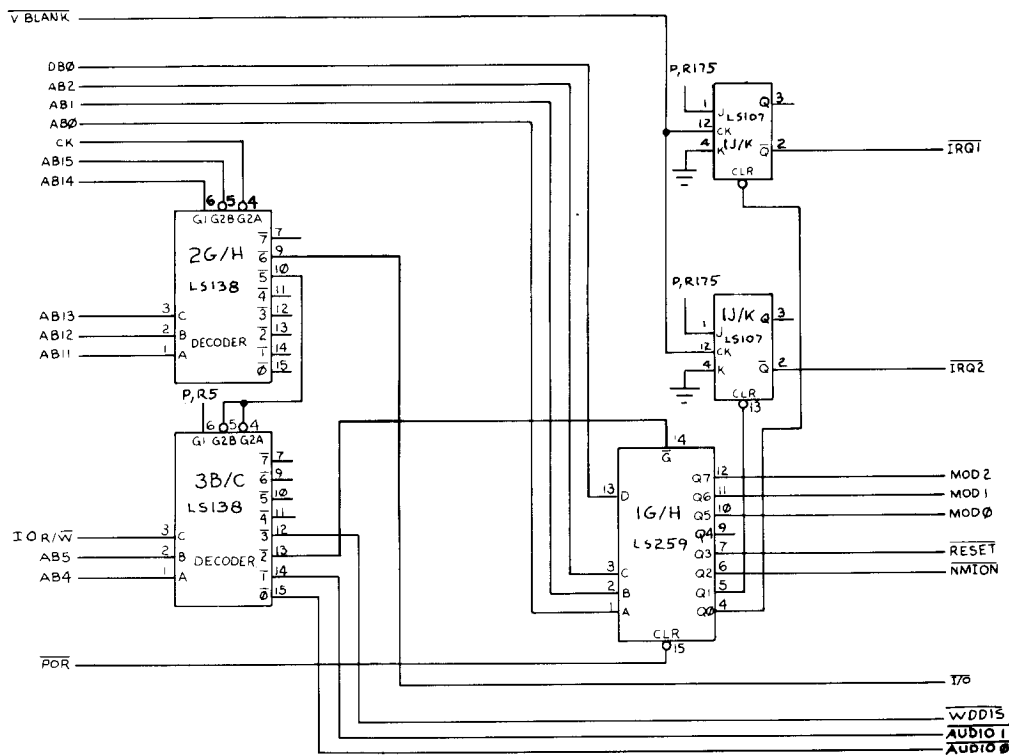
SP-203 Sheet 3B
 1st printing 3L

1st Priority CPU





CPU PCB Address Decoder



NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
 CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

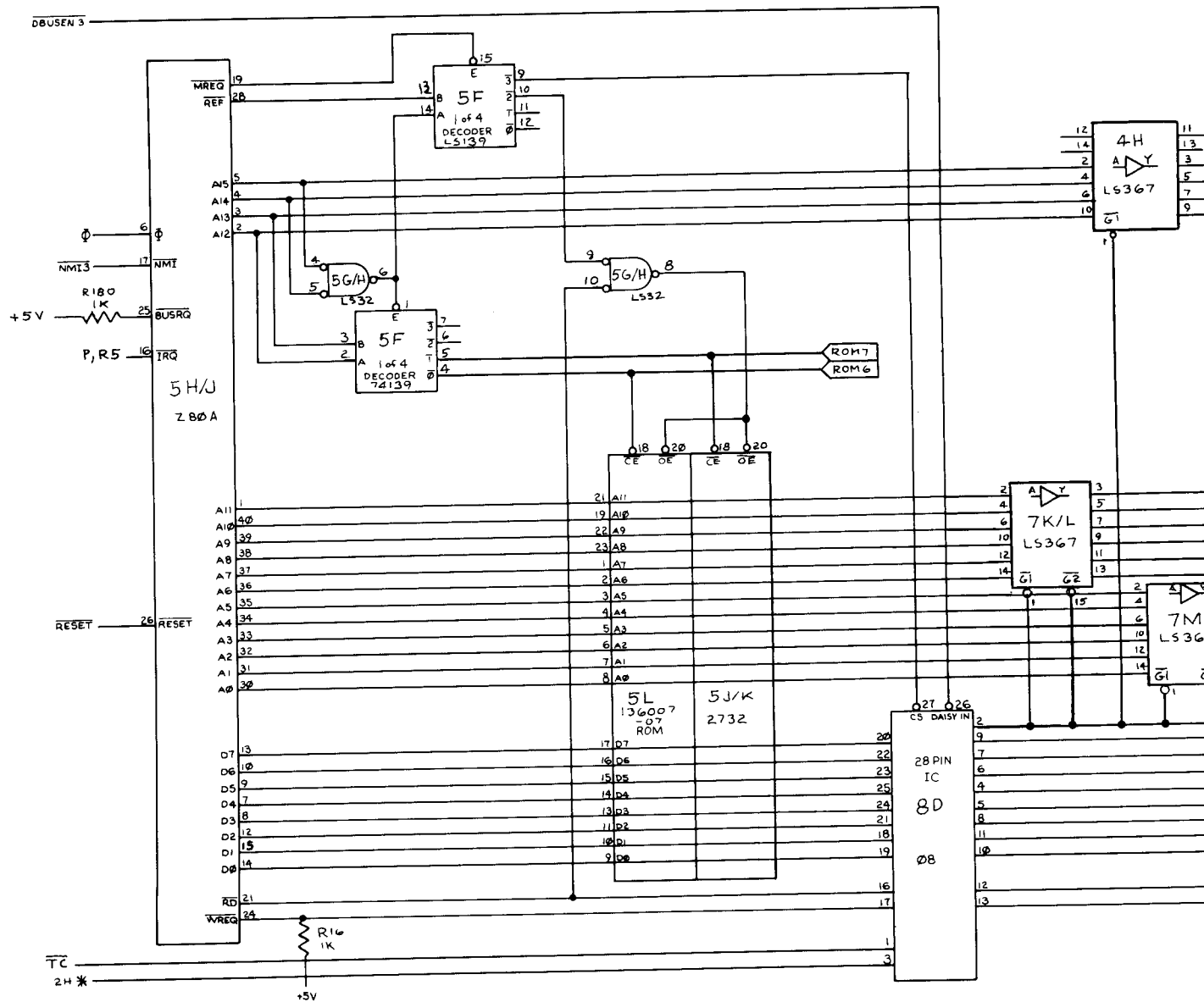
Dig Dug CPU PCB Schematic Diagram

© ATARI INC., 1982

A Warner Communications Company

SP-203 Sheet 4A
 1st printing 3L

3rd Priority CPU



[illegible]

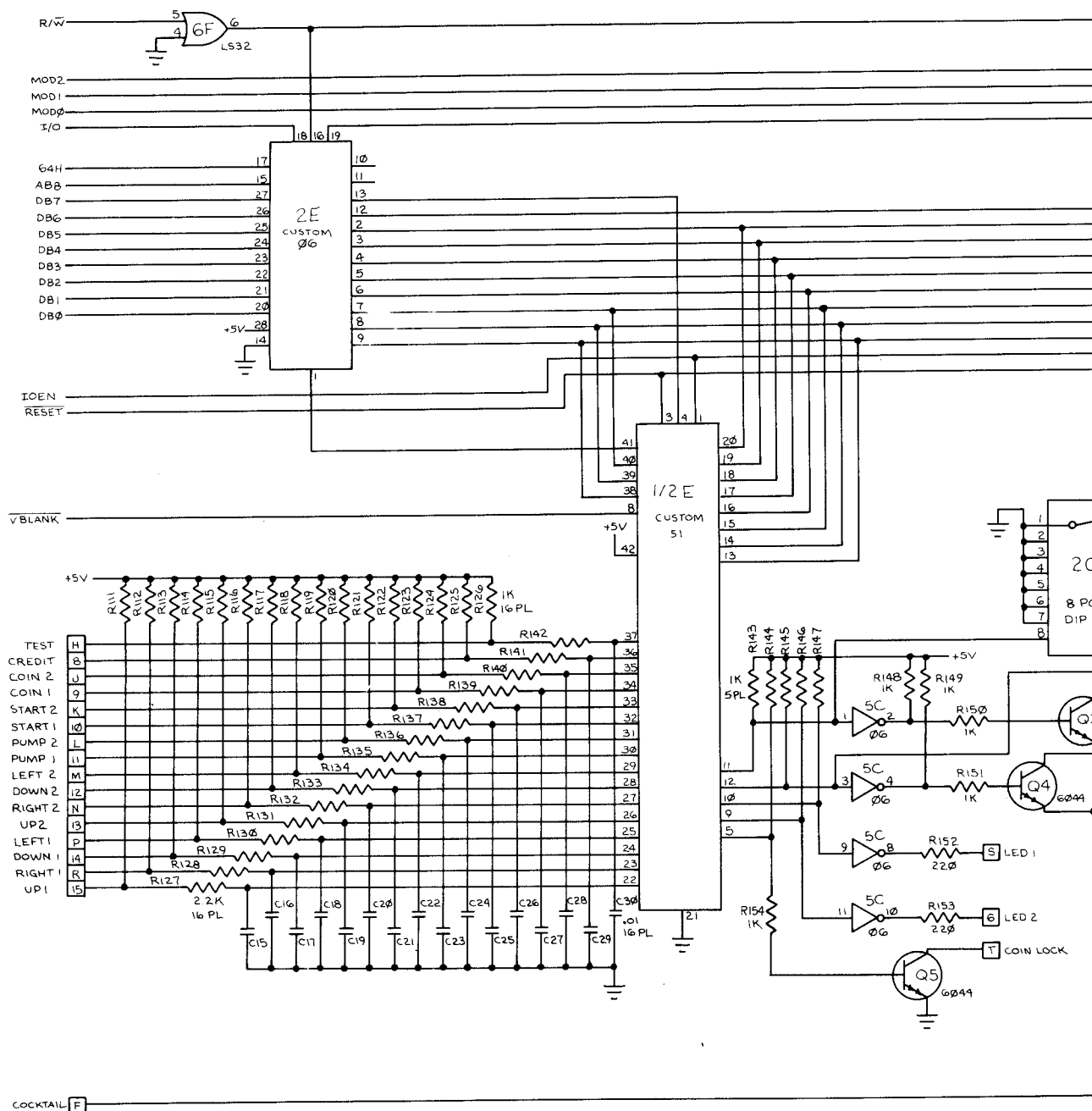
NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

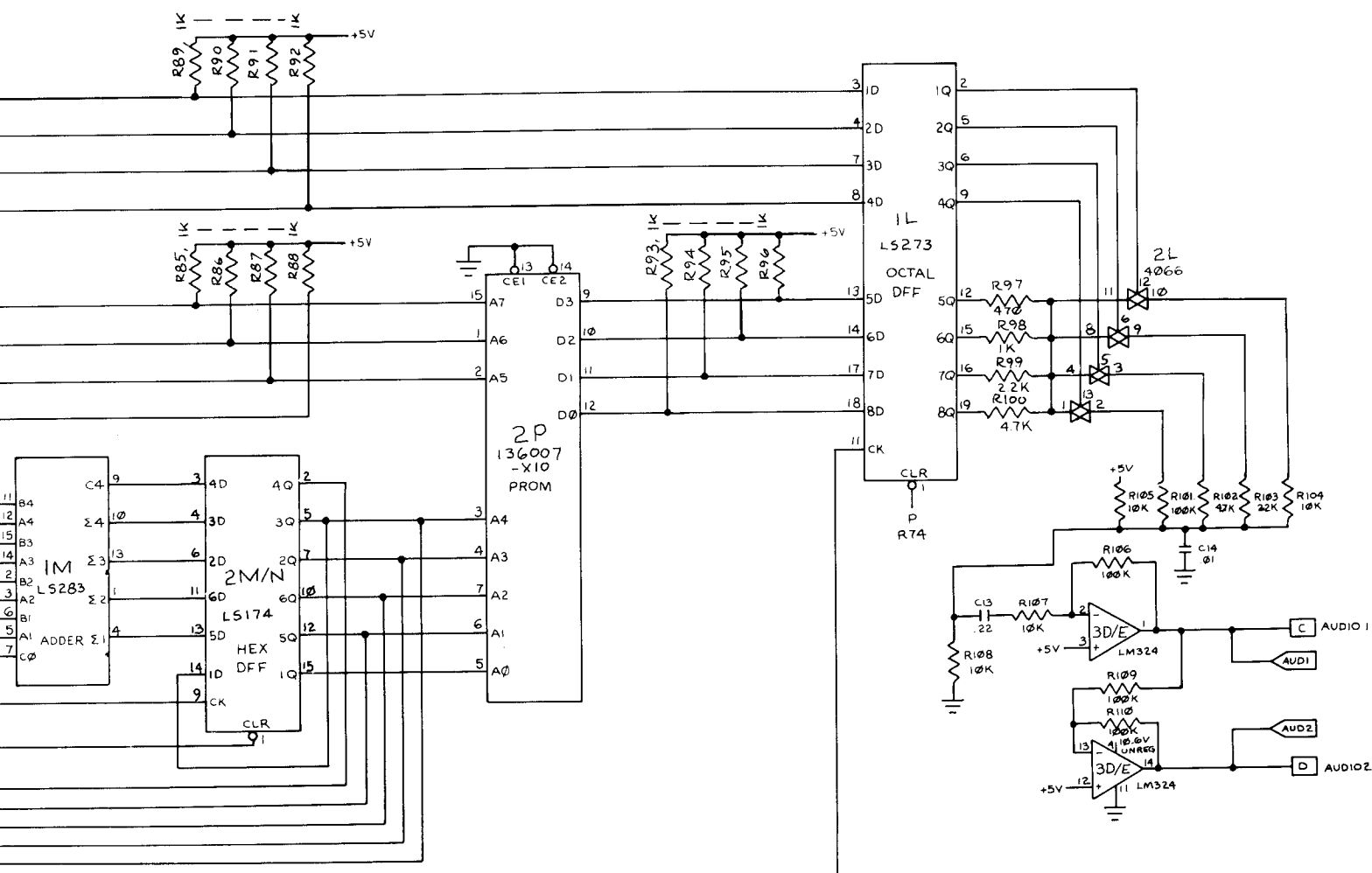
© ATARI INC., 1982



A Warner Communications Company

Switch Inputs, Coin Door and Control P





I/O 2H*



NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
 CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

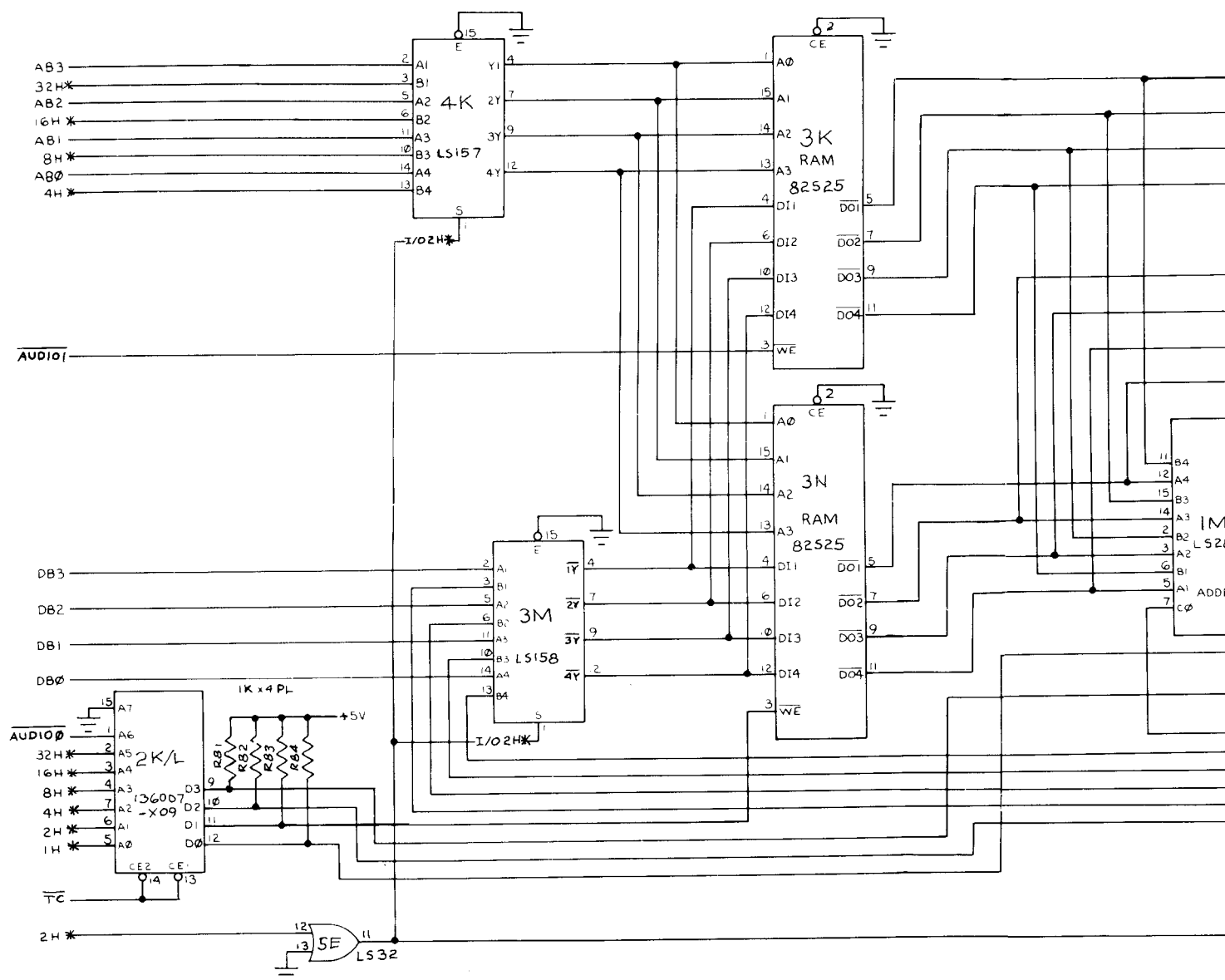
Dig Dug CPU PCB Schematic Diagram

© ATARI INC., 1982

A Warner Communications Company

SP-203 Sheet 5A
 1st printing 3L

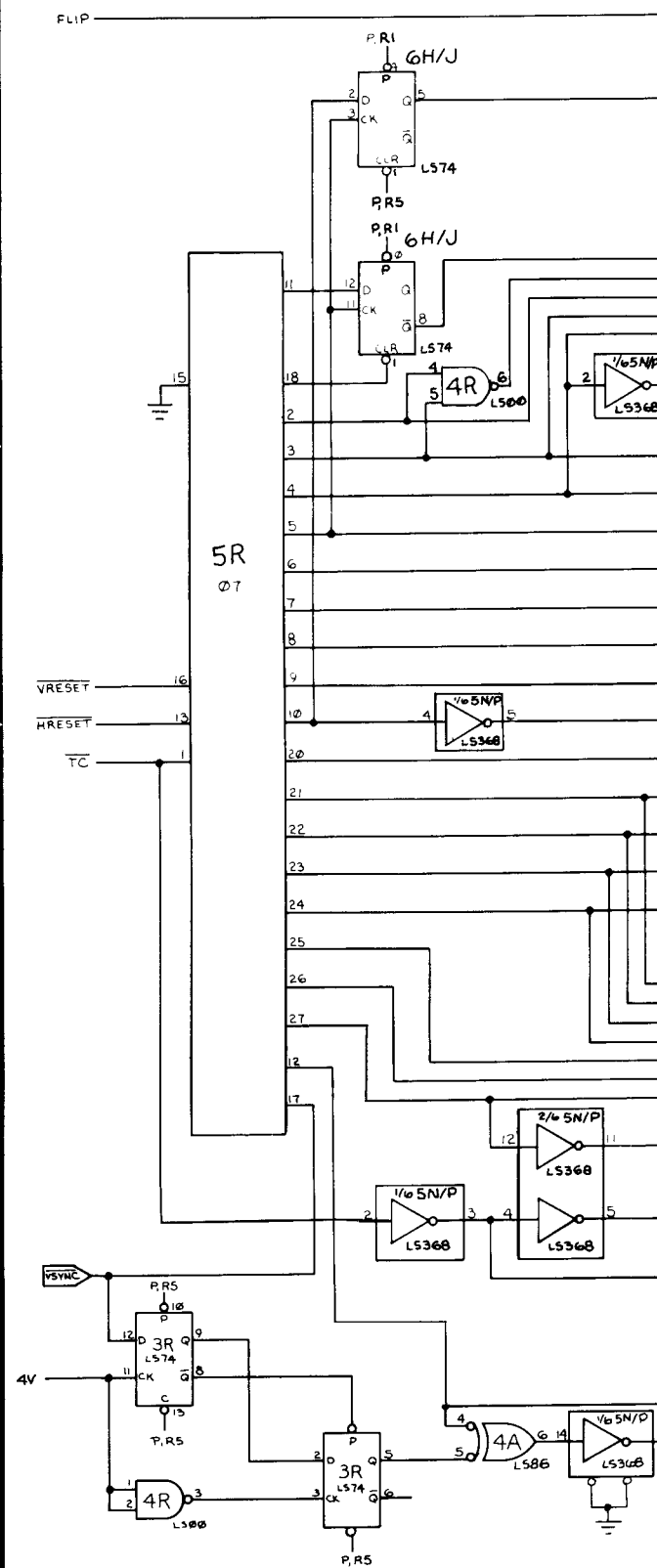
Audio

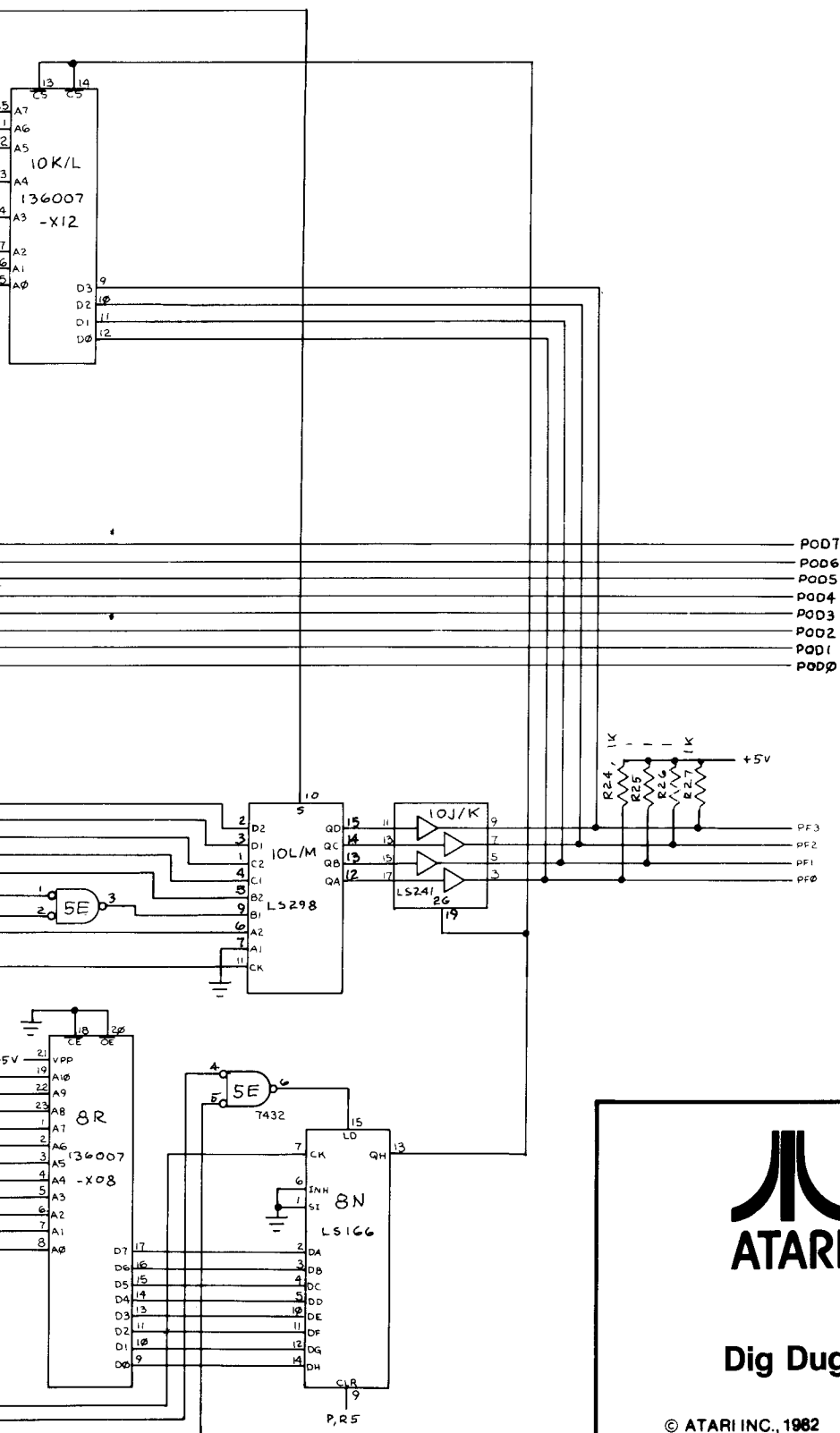




SP-203 Sheet 5B
1st printing 3L

Sync Generator





NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
 CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

Dig Dug Video PCB Schematic Diagram

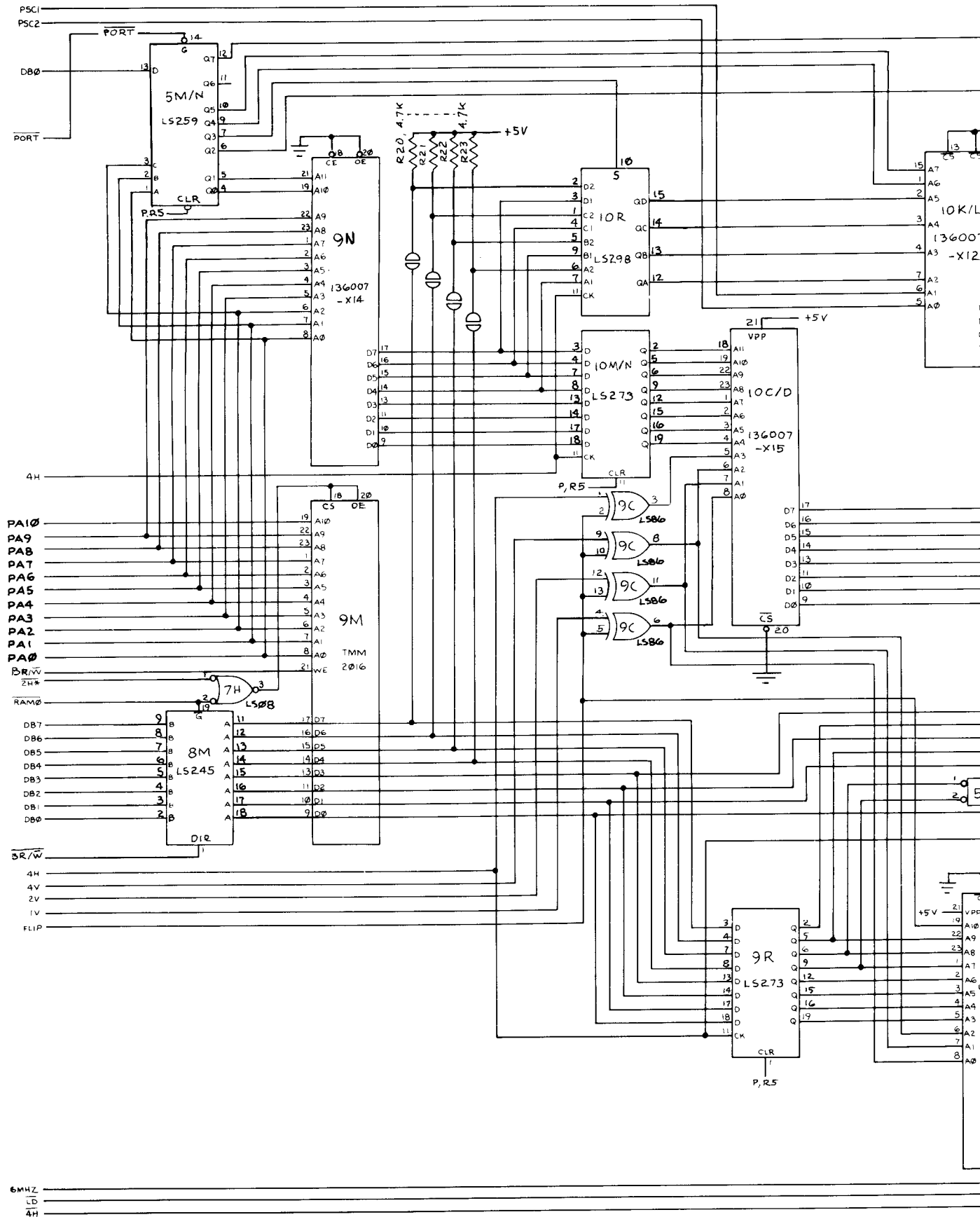
© ATARI INC., 1982



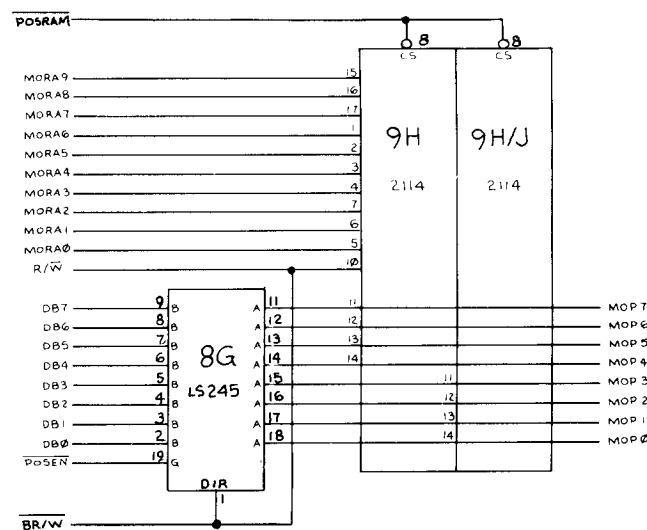
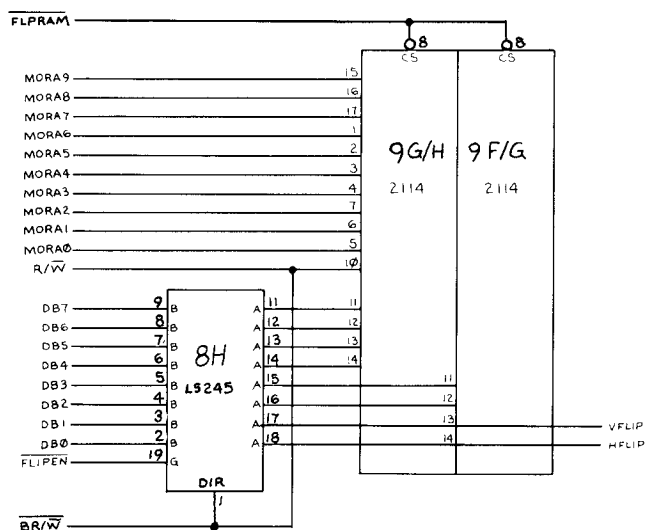
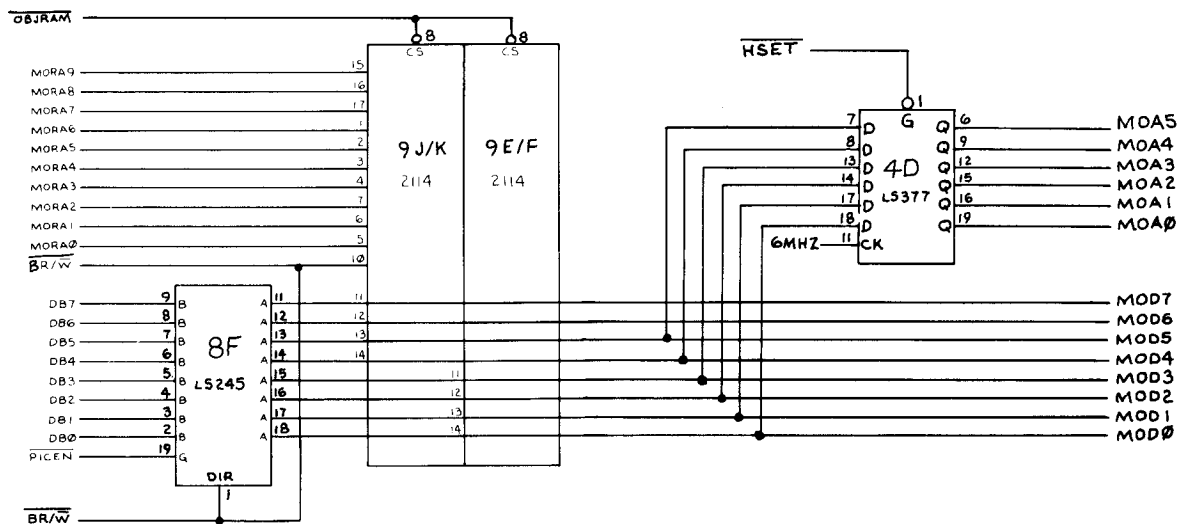
A Warner Communications Company

SP-203 Sheet 6A
1st printing 3L

Playfield Generator



Motion Object RAM



NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
 CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

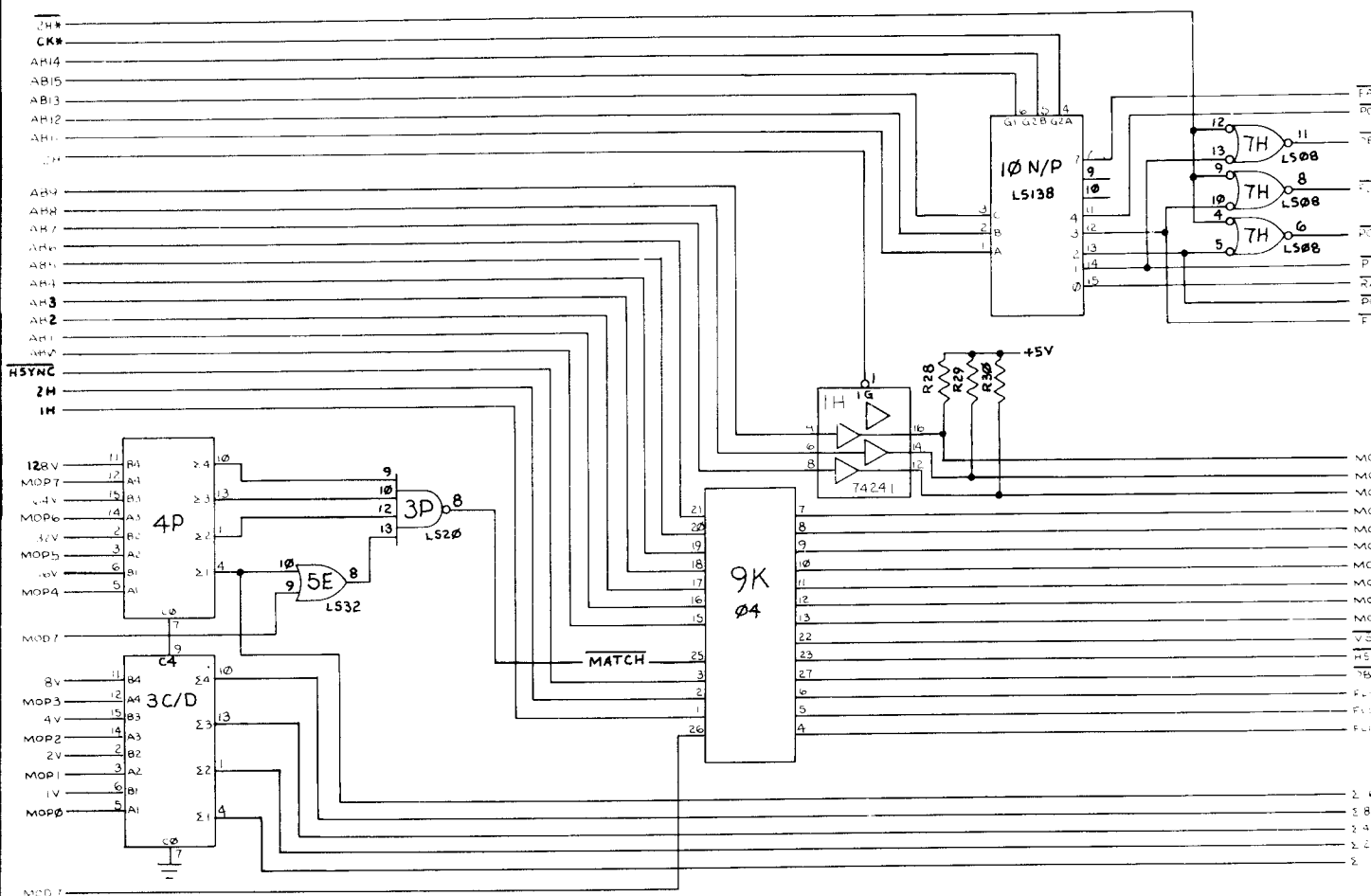
Dig Dug Video PCB Schematic Diagram

© ATARI INC., 1982

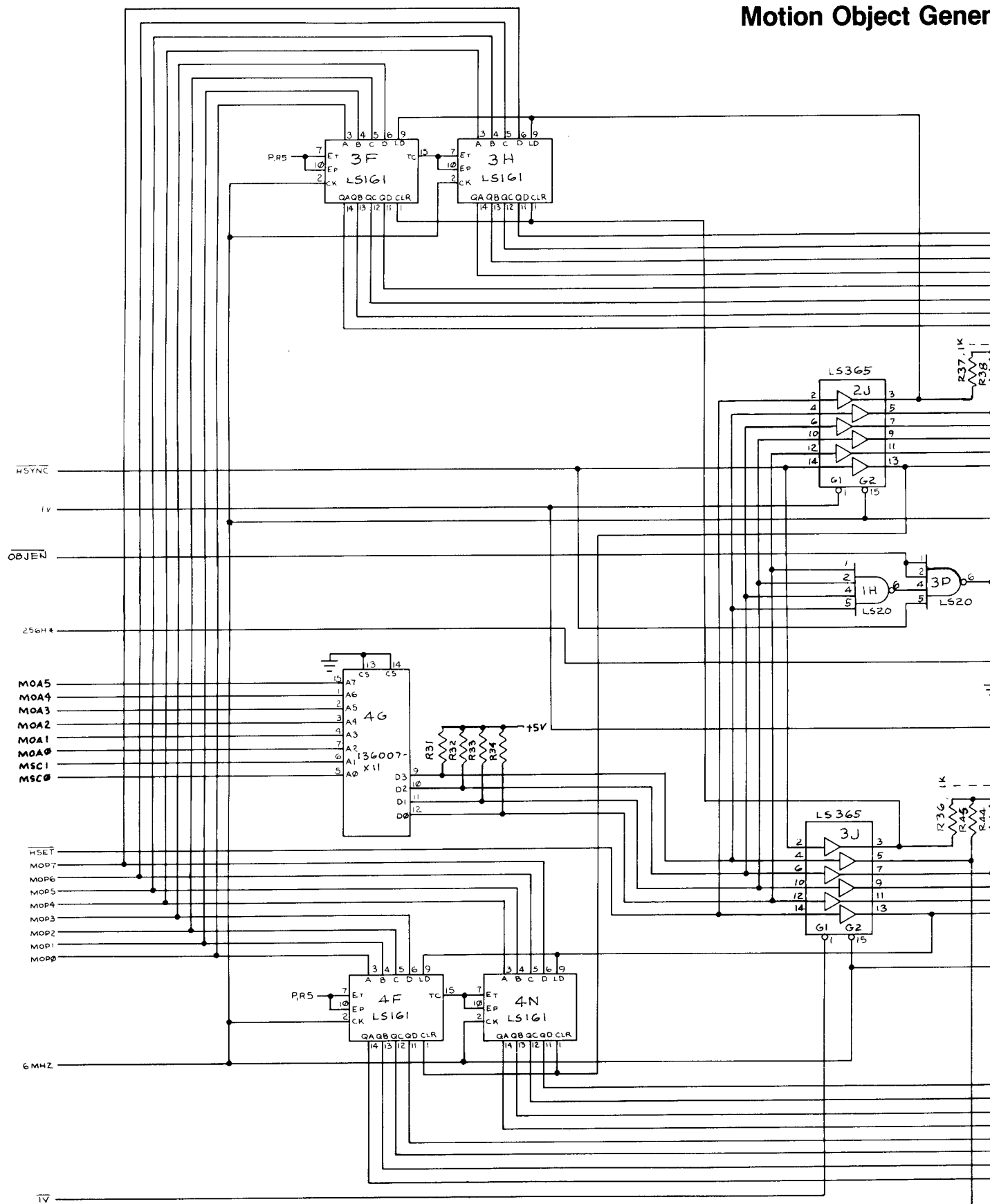
A Warner Communications Company

SP-203 Sheet 6B
 1st printing 3L

Motion Object Address Generator, Decoder, and Match Line Flag

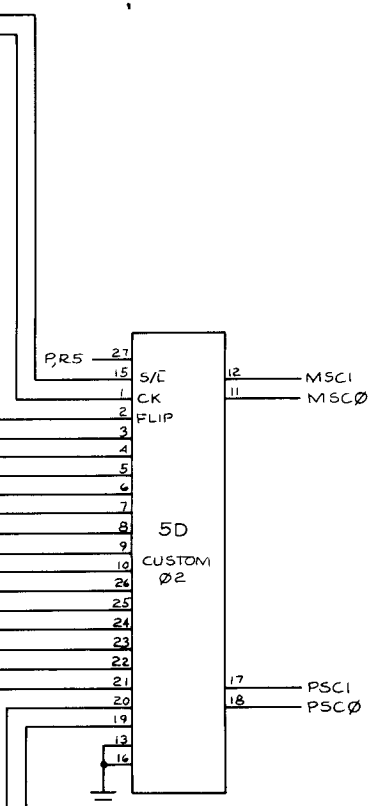
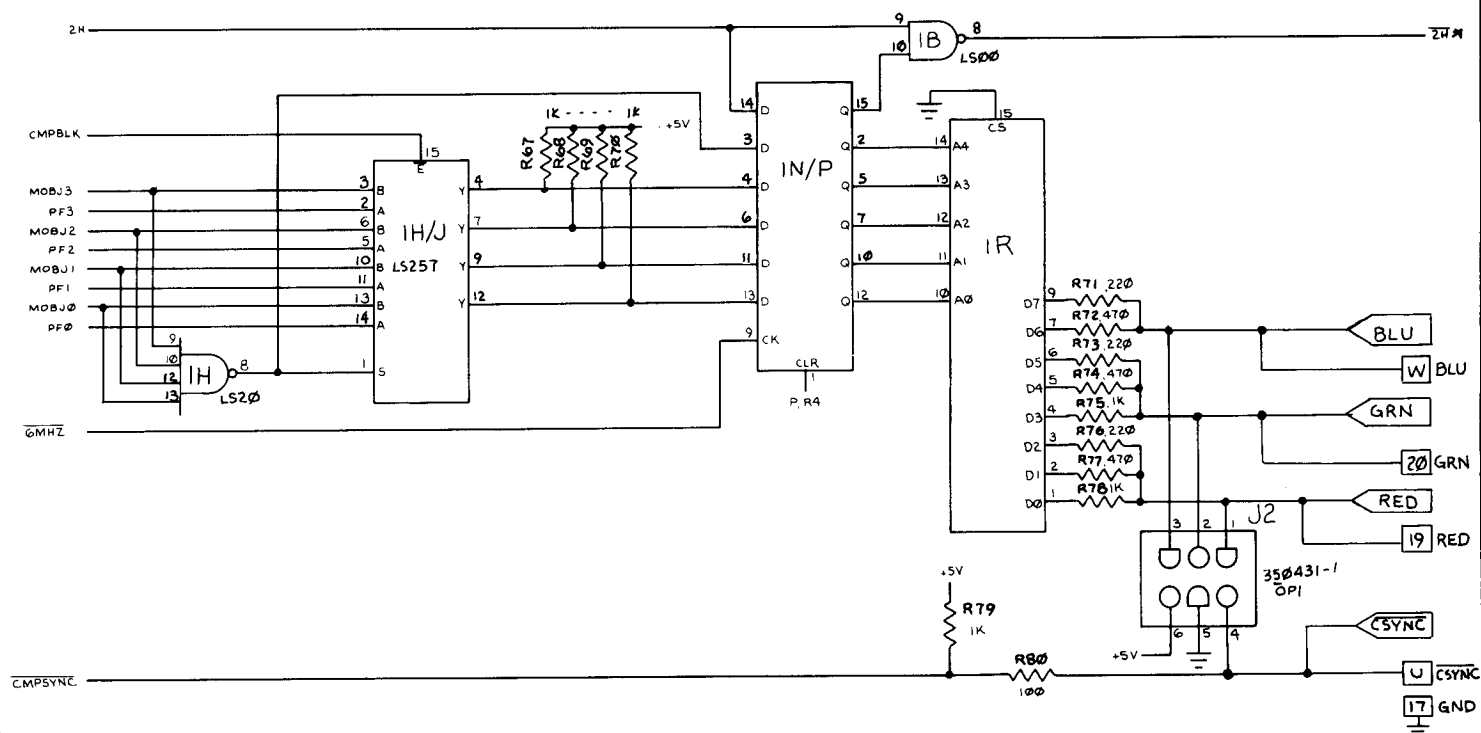


Motion Object Gener



MOP0 through MOP7 - Motion Object
V and H Position Data.
MOBJA0 through MOBJA7 - Motion
Object Picture Address.

Video Output



NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

Dig Dug Video PCB Schematic Diagram

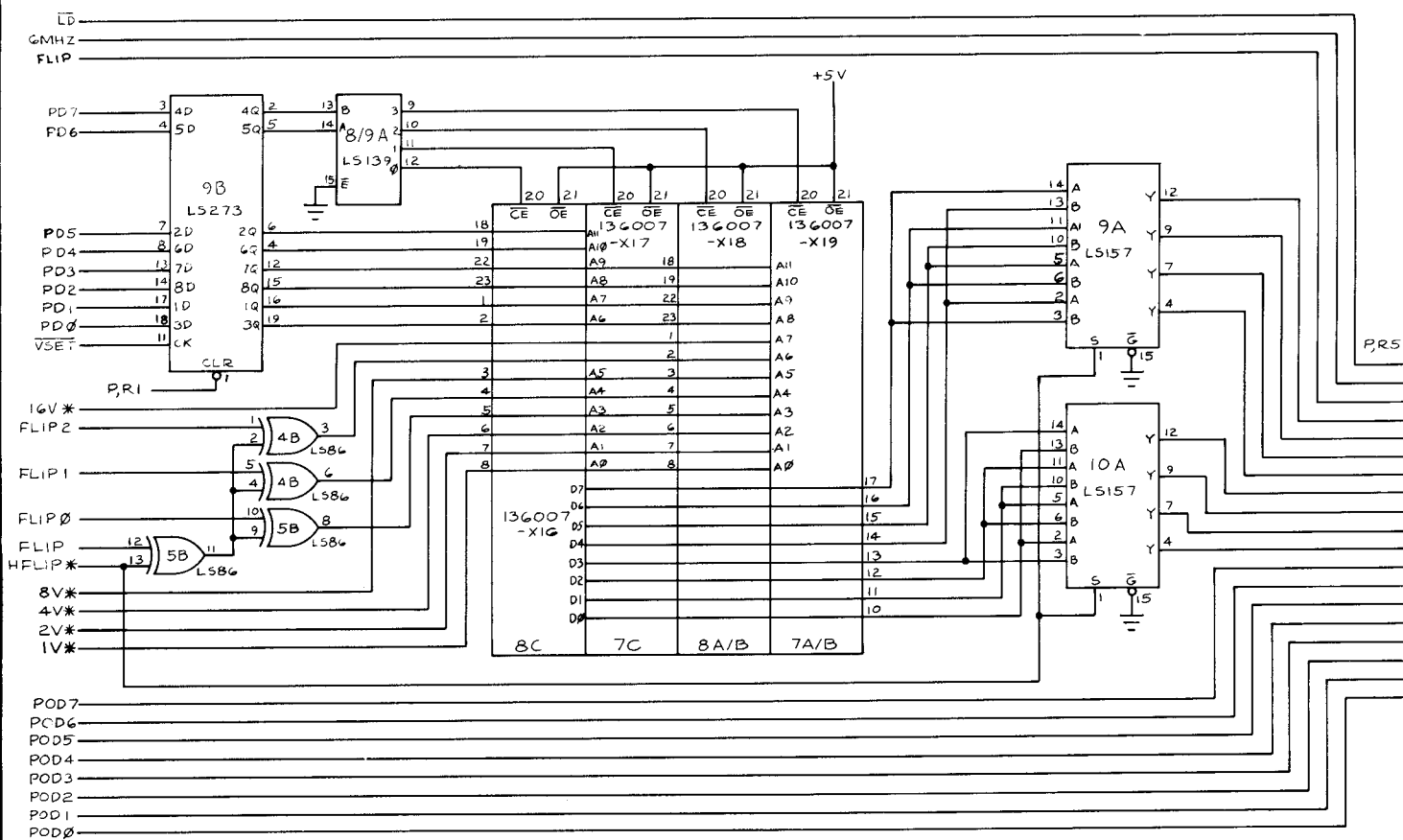
© ATARI INC., 1982



 A Warner Communications Company

SP-203 Sheet 7B
1st printing 3L

Video Color Controller





NOTICE TO ALL PERSONS RECEIVING THIS DRAWING
CONFIDENTIAL: Reproduction forbidden without the specific written permission of Atari, Inc., Sunnyvale, CA. This drawing is only conditionally issued, and neither receipt nor possession thereof confers or transfers any right in, or license to use, the subject matter of the drawing or any design or technical information shown thereon, nor any right to reproduce this drawing or any part thereof. Except for manufacture by vendors of Atari, Inc., and for manufacture under the corporation's written license, no right is granted to reproduce this drawing or the subject matter thereof, unless by written agreement with or written permission from the corporation.

Dig Dug Video PCB Schematic Diagram

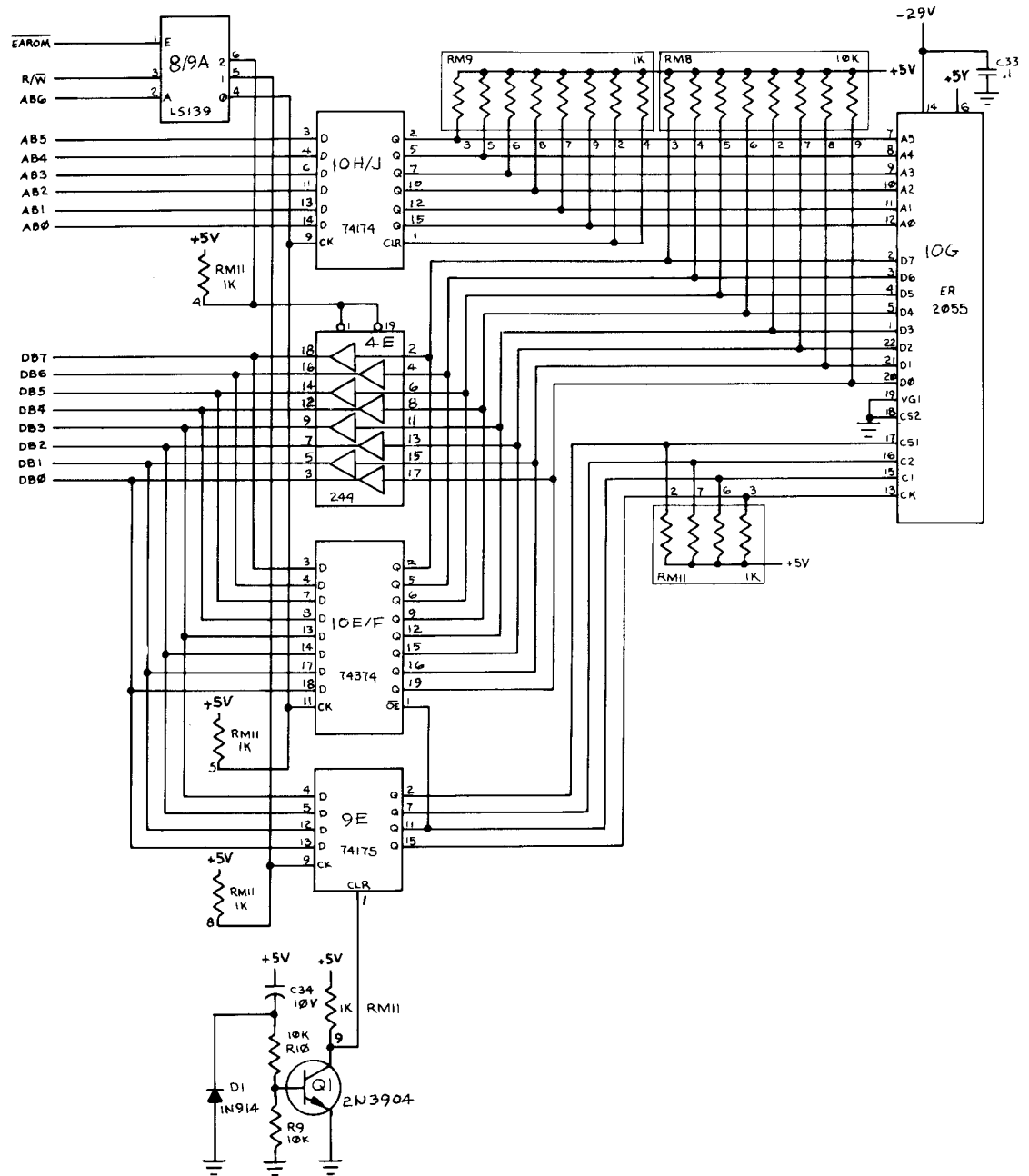
© ATARI INC., 1982



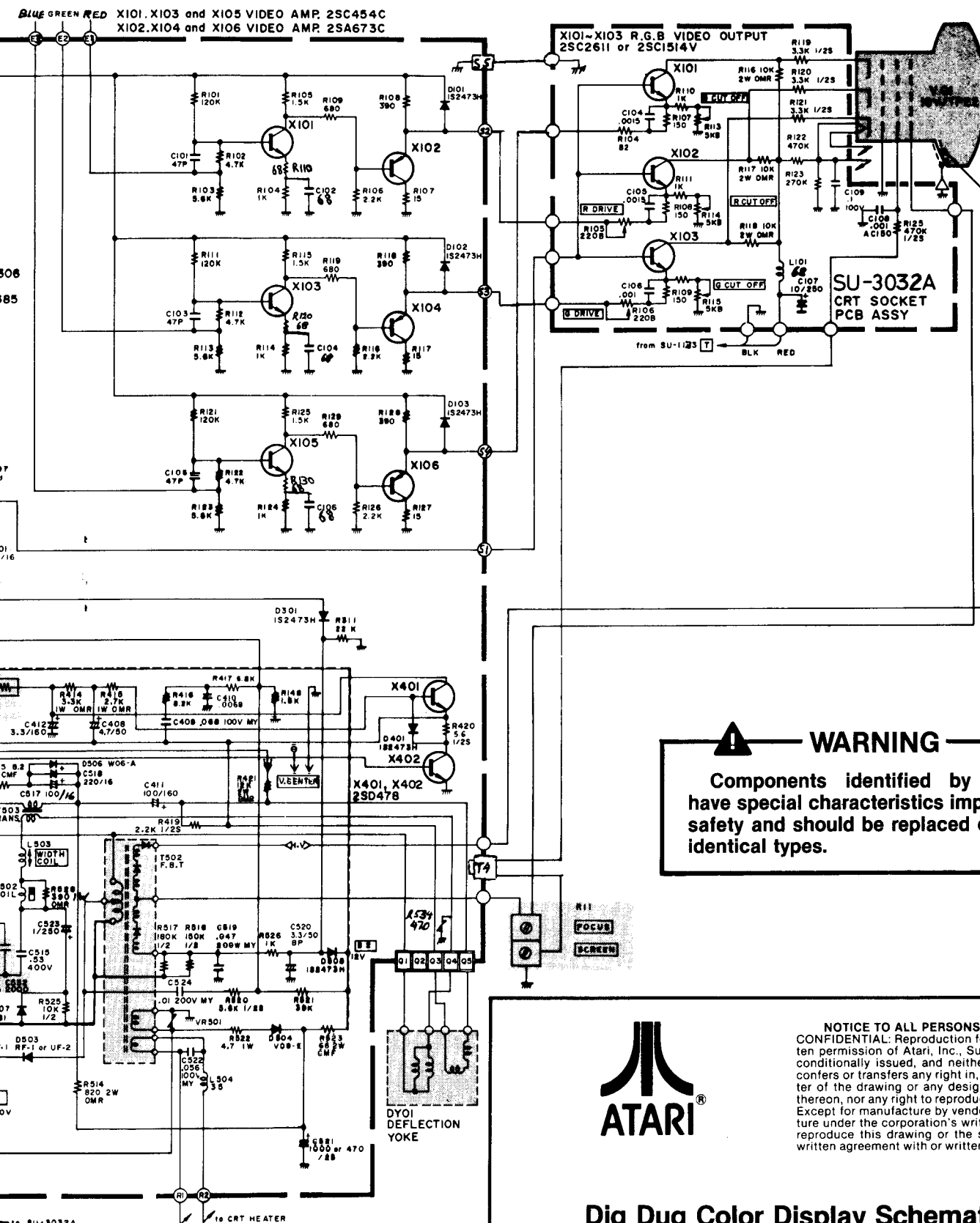
A Warner Communications Company

SP-203 Sheet 8A
1st printing 3L

High-Score Table



Schematic Diagram



Dig Dug Color Display Schematic Diagram

© ATARI INC., 1982

A Warner Communications Company

SP-203 Sheet 8B
1st printing 3L

Schematic Notes

Unless otherwise specified

Resistance: (Ω) (K $\rightarrow$ K Ω , M $\rightarrow$ M Ω), 1/4 (W) carbon resistor

Capacitance: 1 or higher $\rightarrow$ (pF), less than 1 $\rightarrow$ (μ F)

working voltage $\rightarrow$ 50 (V)

ceramic capacitor

Inductance: (μ H)

Electrolytic Cap: Capacitance Value (μ F)/working voltage (V),

NP $\rightarrow$ non-polar (or bipolar) electrolytic cap.

Refer to the parts list for additional component information.

⊙ indicates test point connection

⏏ indicates chassis ground unless otherwise specified

Hz indicates cycles per second

For **safety** purposes (and continuing reliability)

⚠ replace all components marked with safety symbol with identical type.

NOTE: FR $\rightarrow$ fusible resistor ($\text{---}\text{---}\text{---}$)

Parts identification on circuit boards:

e.g. SU1126A (R107 = R1107)

SU3030A (R113 = R3113)

Color Display Scheme

